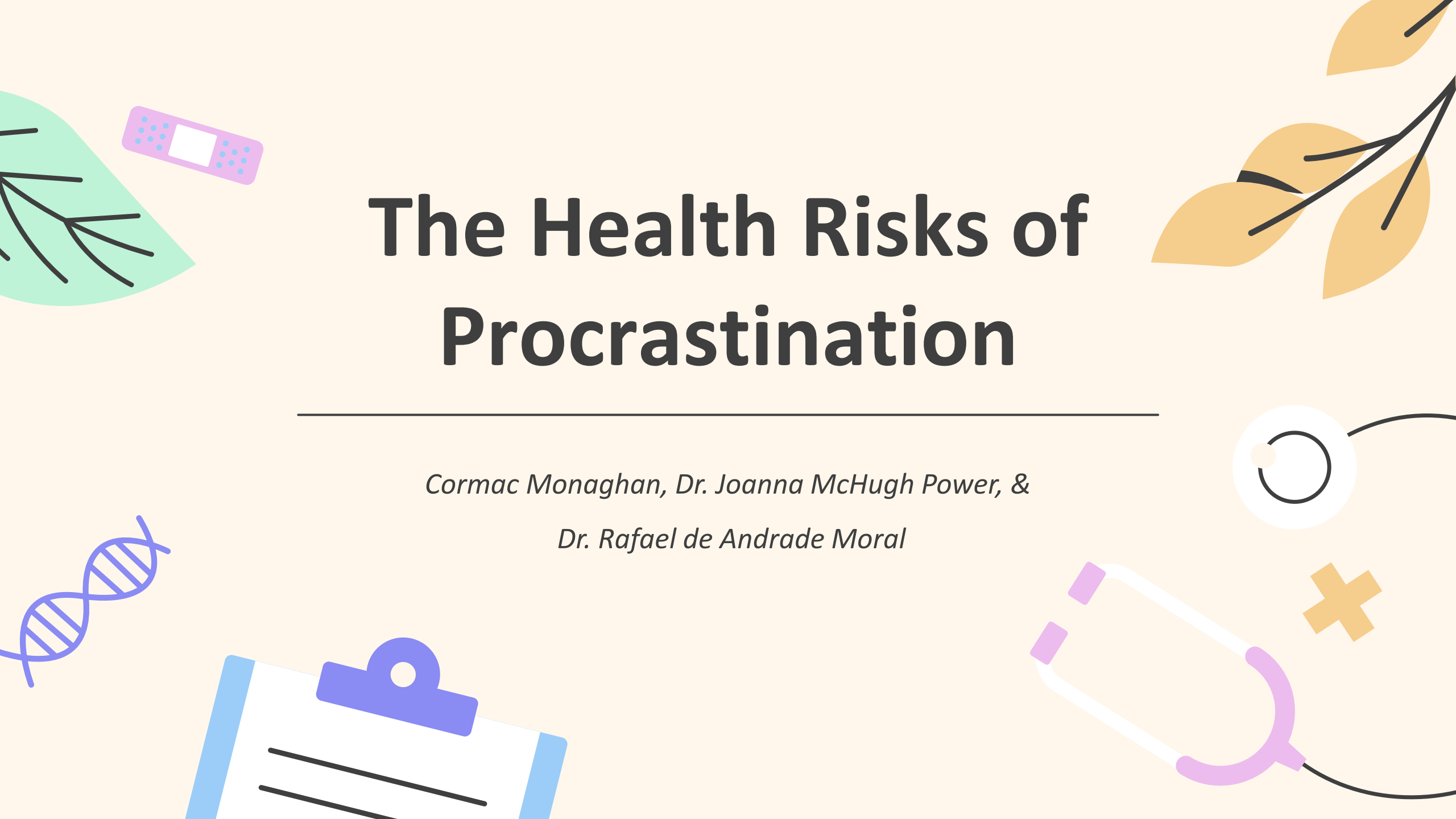




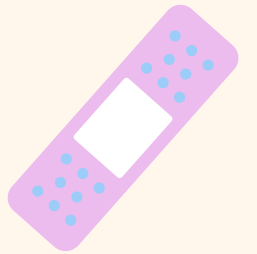
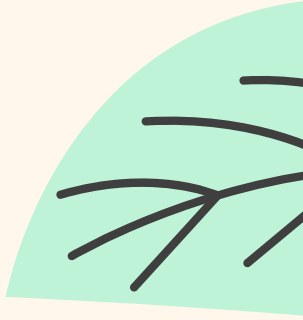
The Health Risks of Procrastination

*Cormac Monaghan, Dr. Joanna McHugh Power, &
Dr. Rafael de Andrade Moral*

Overview

- Procrastination and Health
- Health and Retirement Study (HRS)
- Generalised Additive Models (GAMs)
- Results
- Conclusions

What is Procrastination





Procrastination



To procrastinate is to voluntarily delay an intended course of action despite expecting to be worse off for the delay

Voluntary

Typically, not involuntary, but rather a conscious decision

Intended course of action

Occurs when you have a specific task or goal in mind that you plan to undertake

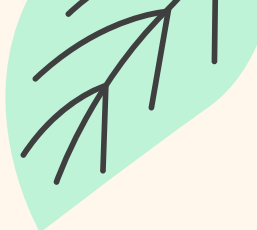
Worse off for the delay

Anticipating that this delay will have negative consequences





Procrastination and Health



Sleep
Quality

ORIGINAL ARTICLE

YOUNG AND OLD ADULTS' HEALTH-RELATED PROCRASTINATION, QUALITY OF SLEEP AND MENTAL WELLBEING

Ghaliyah Firdaus, Saadia Aziz, Saba Akhtar, Saman Sunny

Department of Applied Psychology, National University of Modern Languages, Islamabad, Pakistan

Dental
Health

Original Article

J Epidemiol 2022;32(10):464-468

Association Between Procrastination in Childhood and the Number of Remaining Teeth in Japanese Older Adults

Moemi Shimamura, Yusuke Matsuyama, Ayako Morita, and Takeo Fujiwara

Department of Global Health Promotion, Tokyo Medical and Dental University, Tokyo, Japan

Received August 12, 2020; accepted March 19, 2021; released online April 10, 2021



Procrastination and Health

No study has yet investigated the impact of procrastination on the **general health** and **health behaviours** of older adults

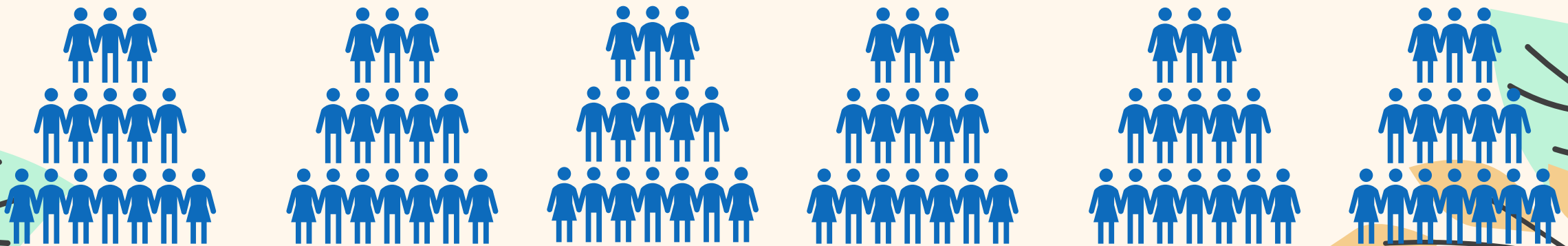
Why is this important

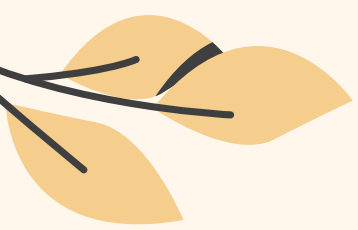
- Older adults are more likely to encounter situations that require them to make important decisions about their health.
- Procrastinating such decisions can have a significant effect on their lives.
- For example, when older adults delay preventive health checkups, they may face more severe outcomes than younger adults.
- Age-related health conditions can worsen more quickly and have more serious implications for older individuals.
- Aging increases the susceptibility to various health problems and diseases and the body's ability to heal and respond to treatments against these problems may decline with age.

Health and Retirement Study

Health and Retirement Study

- The HRS is a longitudinal panel study that surveys approximately 20,000 American adults (predominantly focuses on adults over the age of 50).
- Every 2 years detailed information pertaining to respondent's physical, cognitive, financial, employment, family, and psychosocial conditions is collected.

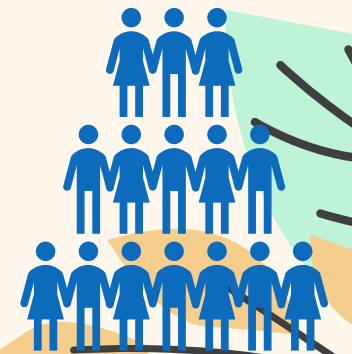
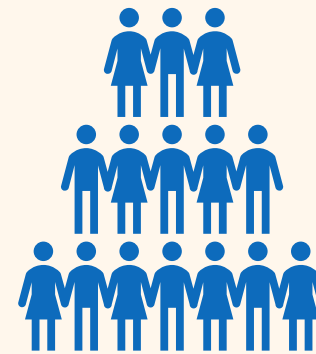
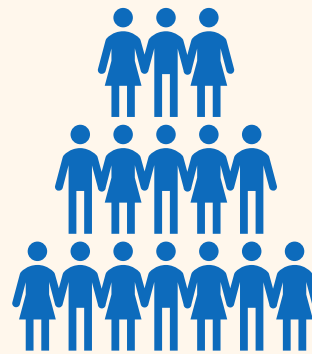
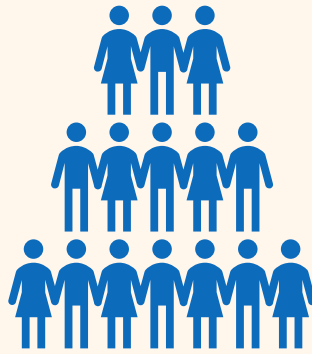
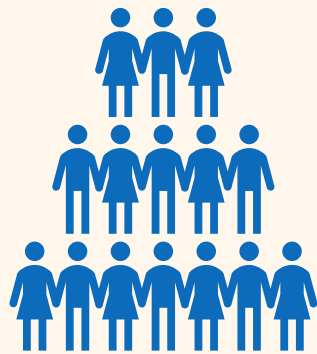
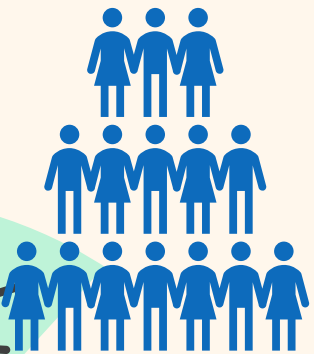




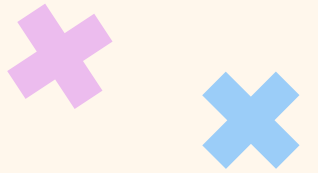
Health and Retirement Study



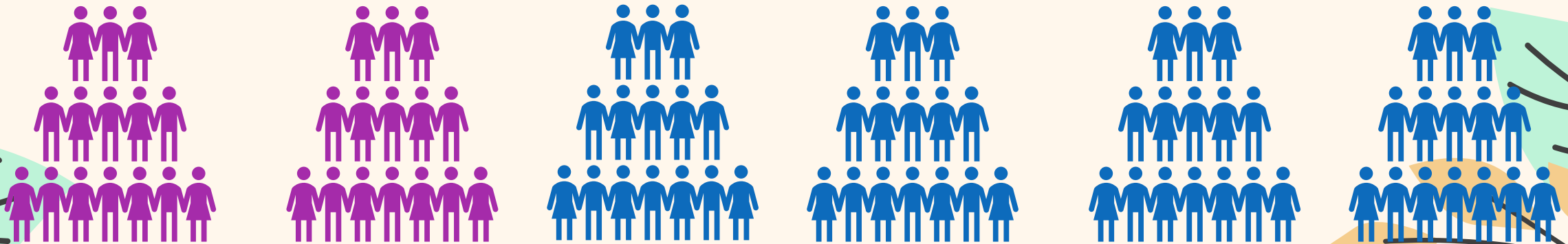
- Each wave includes **experimental modules** that are administered at the end of the core interview.



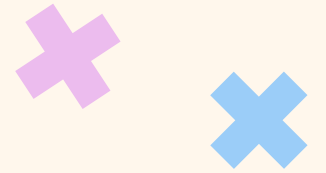
Health and Retirement Study



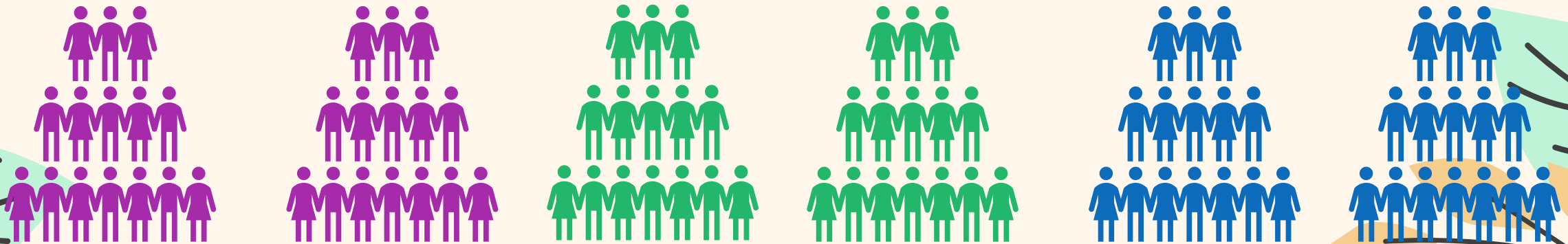
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Health and Retirement Study



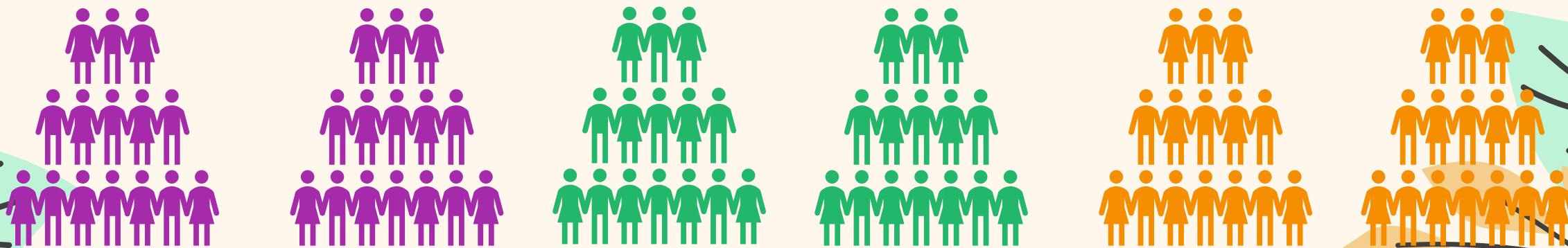
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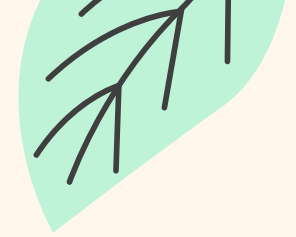
Health and Retirement Study

- Each wave includes **experimental modules** that are administered at the end of the core interview.
- In the 2020 wave, one of these experimental modules included a measure of procrastination.

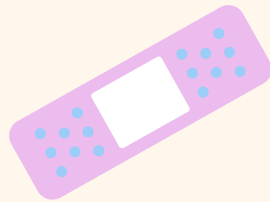
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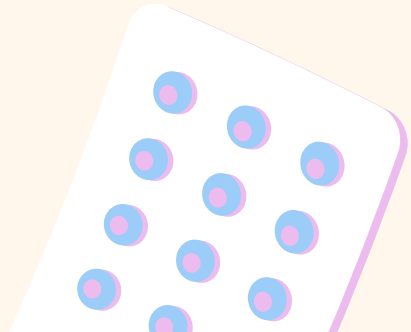


Health Data

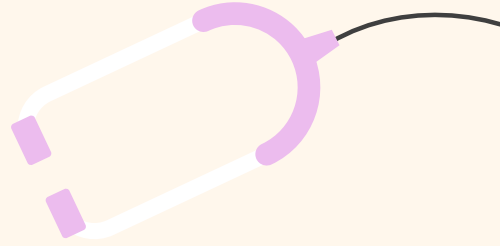


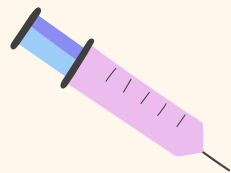
Health Problems



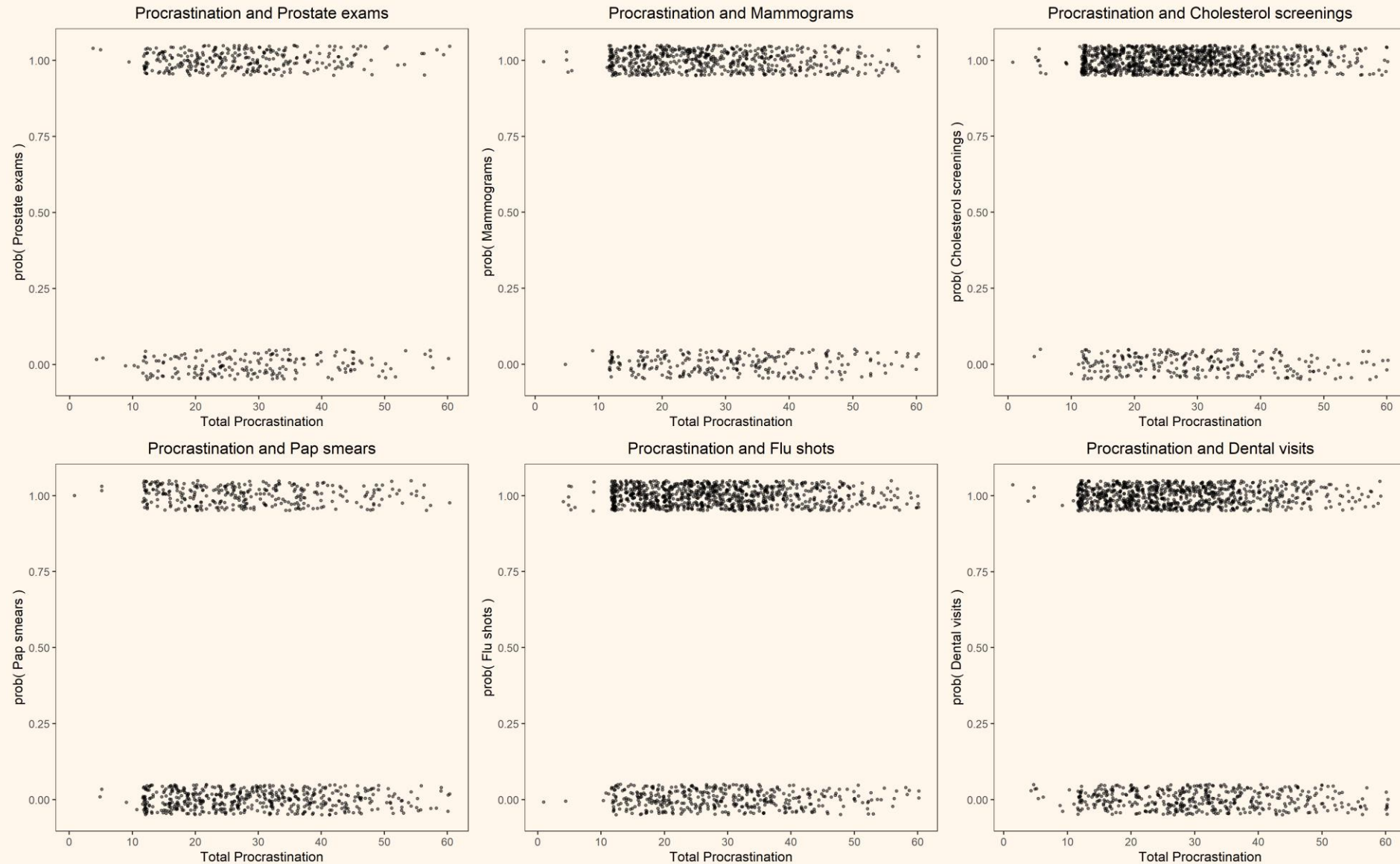
1. Back Pain
 2. Fatigue
 3. Headaches
 4. Diabetes
 5. Cholesterol
 6. Heart Condition
 7. Blood Pressure
 8. Smoking
- 

Health Protection



1. Prostate Exams
 2. Mammograms
 3. Pap Smears
 4. Cholesterol Screenings
 5. Flu Shots
 6. Dental Visits
- 
- 

Health Data - example



Generalised Additive Models

Linear Models

GAMs

Black-Box ML



Generalised Additive Models

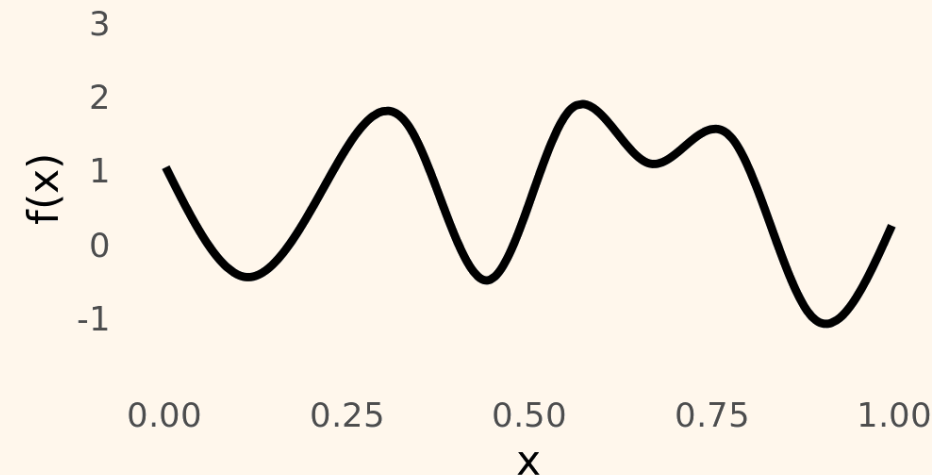
GAMs are effectively a **nonparametric form of regression** where instead of assuming that the response variable depends on a linear combination of predictors:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_j x_j + \epsilon$$

we instead allow the response variable to depend on **smooth functions** of the predictors

$$y = \alpha + s_1(x_1) + s_2(x_2) + \cdots + s_j(x_j) + \epsilon$$

as opposed to having one fixed coordinate β_j , the function s_j can continually change over the range x . The “wiggleness” of s_j is often controlled using penalized regression using either generalized cross validation or restricted maximum likelihood.

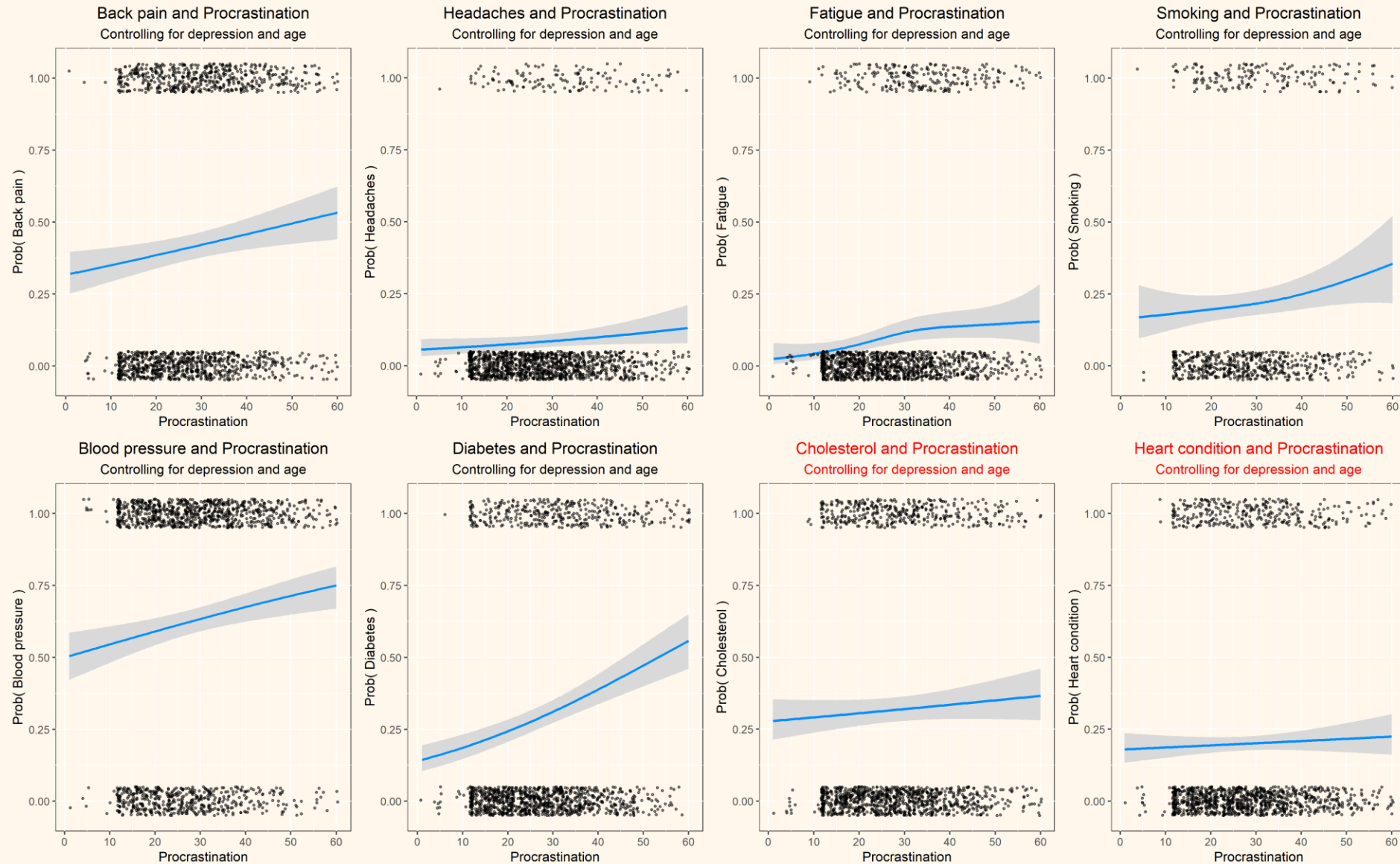




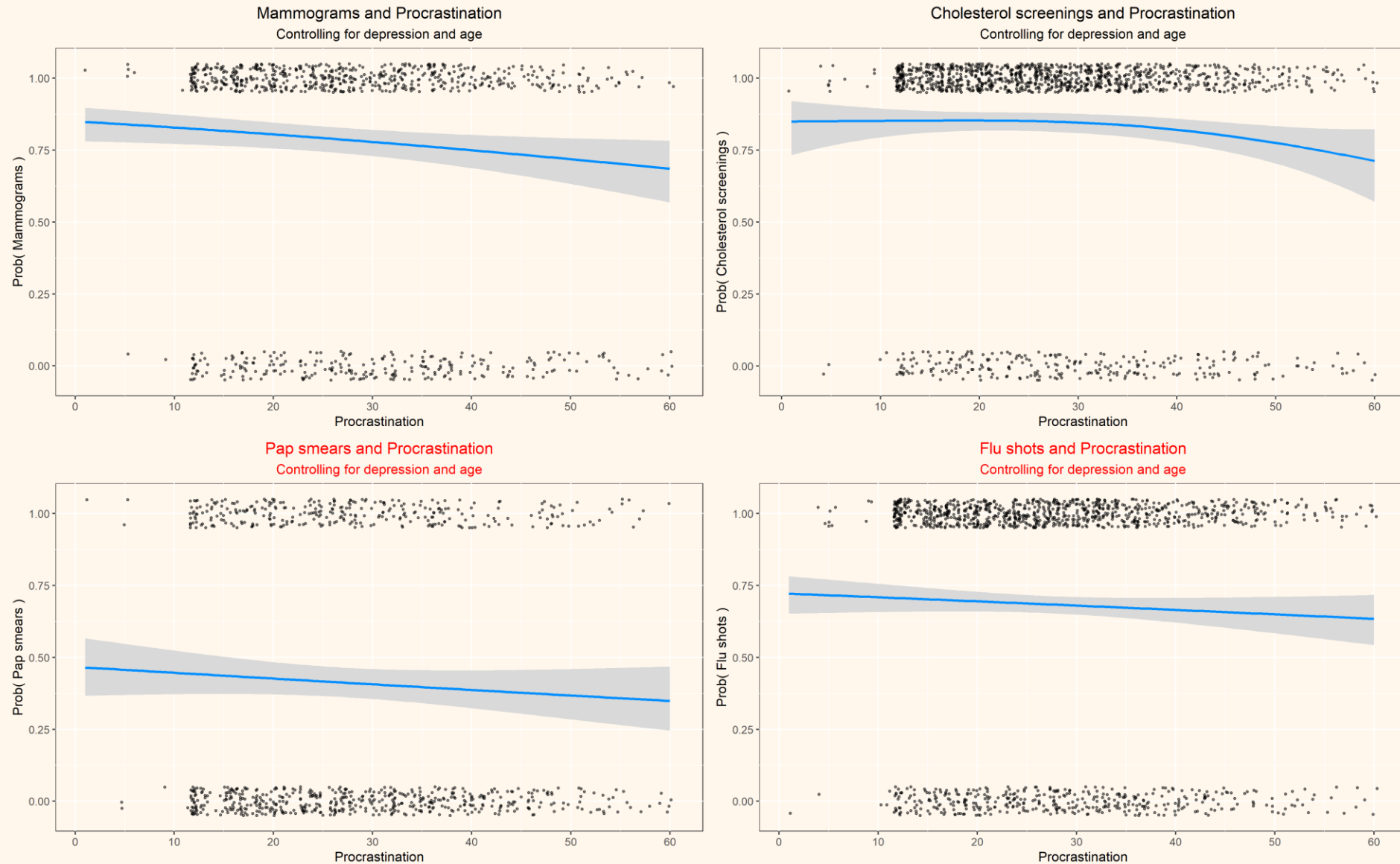
Results

Work in progress

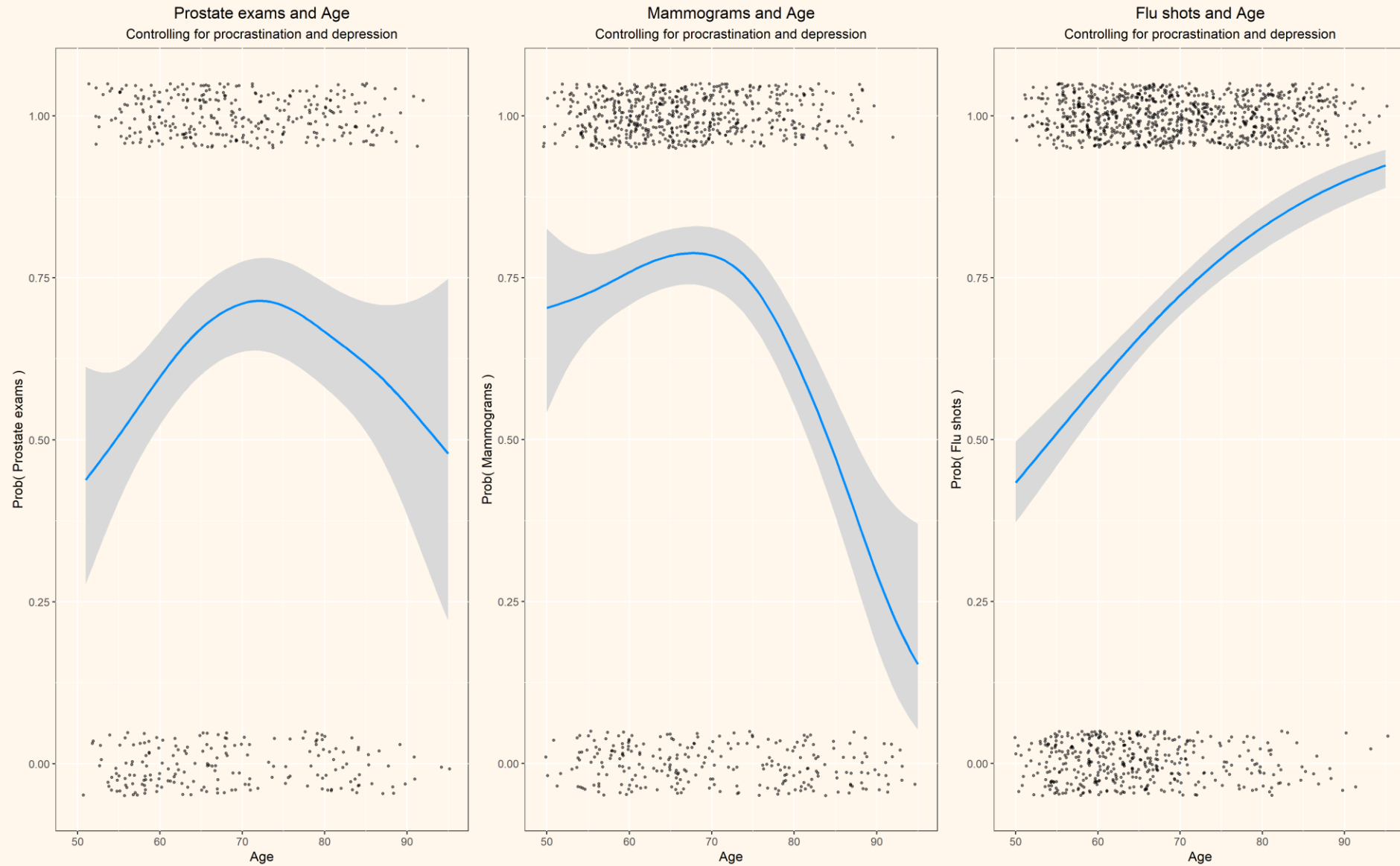
Results: Health Problems



Results: Health Protection



Results: Health Protection





Procrastination and Health



Conclusions

- In older adults, higher levels of procrastination was associated with an increased risk of several health problems.
- Additionally, procrastination was associated with a reduction in self care behaviours.
- Considering that procrastination is still a prevalent behaviour in older adults, these findings may be of importance to further the understanding of older adult health.

Current/Future Work

Interaction Effects

Modelling several interaction effects using **tensor product smooths**. Unlike standard smooths $s_j(x_j)$ which capture the non-linear effect of a single predictor, tensor product smooths capture the non-linear joint effects of two predictors.

$$y = \alpha + \sum_j s_j(x_j) + s_t(x_m, x_n) + \epsilon$$

Tensor product
smooth

Need a good way to
visualise these interactions

Modelling Alcohol Consumption

Previous research suggests that procrastination is also associated with an increase in alcohol consumption. However, data related to alcohol is often tricky to work with.

- Heavily zero inflated
- J-shaped curves
- Social desirability

Attempting to model with the “*gamlss*” package

Thank You

